
A Bunched Approach to Directed HoTT

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Outline

- ① Homotopy type theory
- ② Higher categories
- ③ Simplicial type theory
- ④ The problem of hom types
- ⑤ The bunched solution
- ⑥ From theory to application

Presentation

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- Mathematical language (like Set Theory)
- Constructive flavor ($\neq$ Classical Set Theory)
- Types behaves as spaces.
- Synthetic tool for higher algebra, homotopy theory...
- Suitable for formalizing mathematics / homotopy theory
- **Goal:** replace spaces by *directed spaces*

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- Statements have the form $\Gamma \vdash a : A$ or $\Gamma \vdash a \equiv b : A$.
- “Basic types” and “constructors” as building blocks.

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- **natural numbers type**, $\mathbb{N}$.
 $0 : \mathbb{N}$, $\text{succ} : \mathbb{N} \rightarrow \mathbb{N}$.
 $n : \mathbb{N} \rightsquigarrow \text{succ}(n) : \mathbb{N}$.
eg., $\text{succ}(\text{succ}(\text{succ}(0))) : \mathbb{N}$.

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- Sum type: $A + B$

Hierarchy of Universes

- Tarski (or Grothendieck) Universes. $\mathcal{U}$ is the “type of types”
- $\mathcal{U}$ closed under $\times$, $+$ and $\rightarrow$.
- Having $\mathcal{U} : \mathcal{U}$ leads to paradoxes

$$\mathcal{U}_0 : \mathcal{U}_1 : \mathcal{U}_2 : \dots$$

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• $A + B$:

For any C , $A + B \rightarrow C$ is the same as $A \rightarrow C$ and $B \rightarrow C$.

More type formers

- Π -types:

$$A : \mathcal{U}, B : A \rightarrow \mathcal{U} \quad \rightsquigarrow \quad \prod_{a:A} B(a) : \mathcal{U}.$$

$$f : \prod_{a:A} B(a), x : A \quad \rightsquigarrow \quad f(x) : B(x).$$

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- Identity-types:

$$A : \mathcal{U}, x, y : A \quad \rightsquigarrow \quad x =_A y : \mathcal{U}.$$

$$x : A \quad \rightsquigarrow \quad \text{refl}_x : x =_A x.$$

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Correspondance between types and logical propositions.

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Logic $\longleftrightarrow$ Algebra

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- Think of $x = y$ as the type of paths from x to y .

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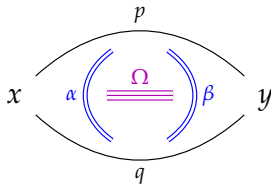
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Theorem

Every type is equipped with a canonical structure of ∞ -groupoid.

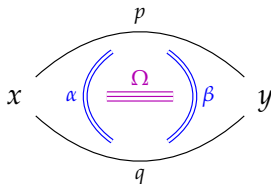
∞ -groupoids

Spaces up to homotopy



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$$x, y : A$$

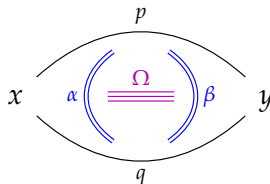
$$p, q : x =_A y$$

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Formalisation of homotopy theory

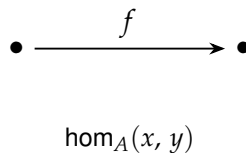
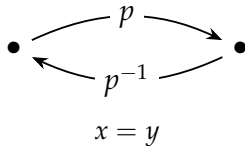
(Feudenthal, long exact sequences, sphere homotopy groups...).

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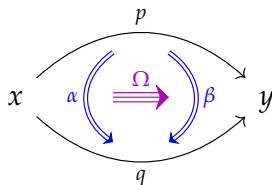


∞ -categories

"Directed" spaces up to homotopy

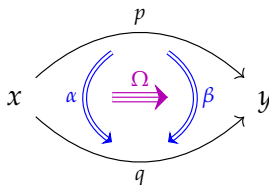
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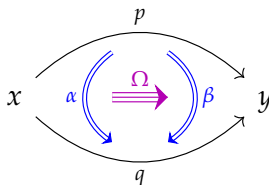
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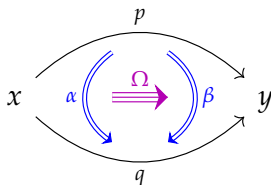
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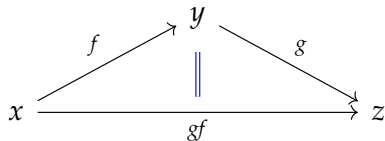
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Rewriting, concurrency,
synthetic manipulation of higher categories.

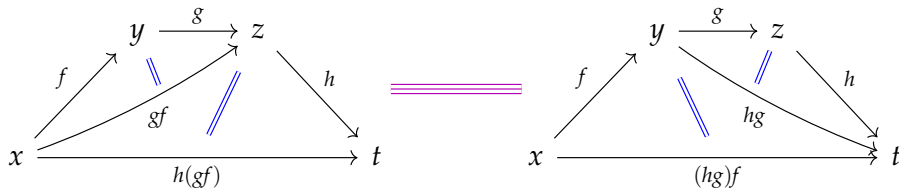
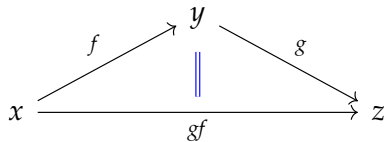
Homotopy coherence

ordinary categories



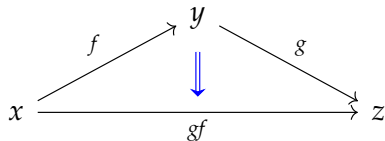
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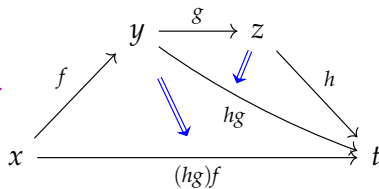
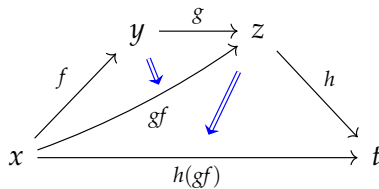
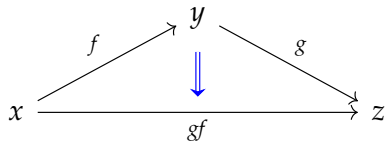
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The nerve construction

Idea:

categories $\leadsto$ simplicial set

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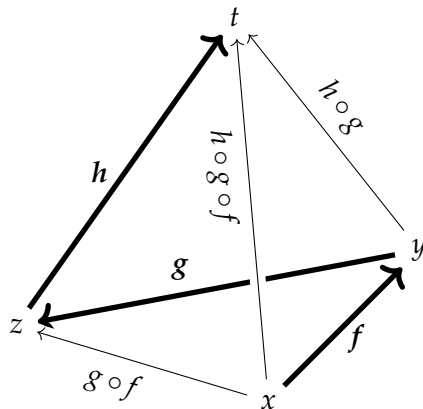
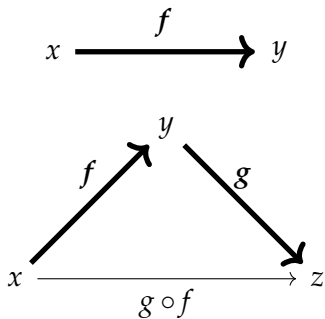
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$$x \xrightarrow{f} y$$

The nerve construction

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Simplicial spaces

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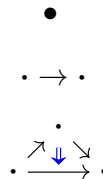
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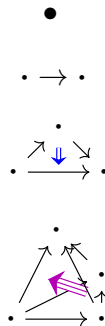
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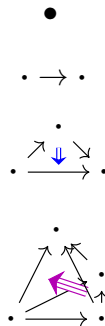


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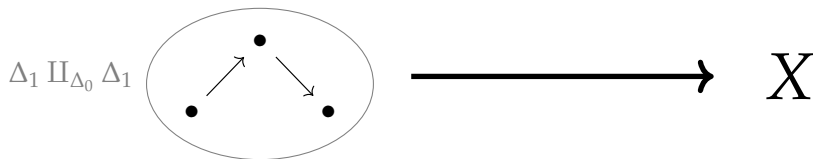
Space valued presheaf $X : \Delta^{\text{op}} \longrightarrow \mathcal{S}$.



Segal spaces

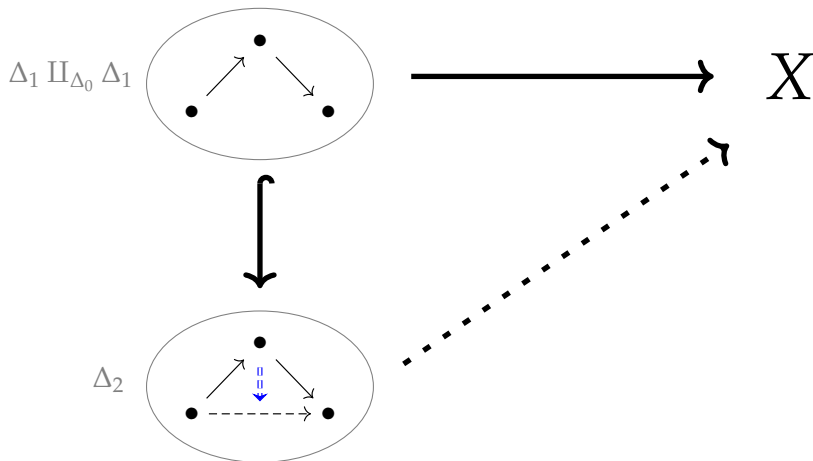
Idea:

encode composability as lifting properties.



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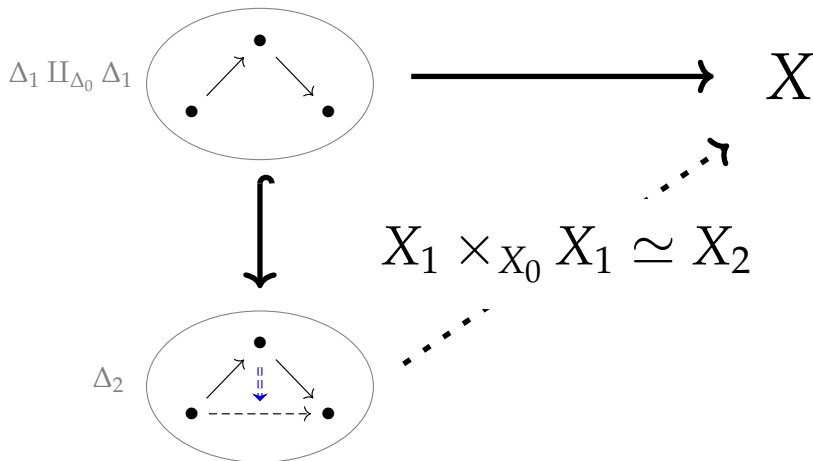
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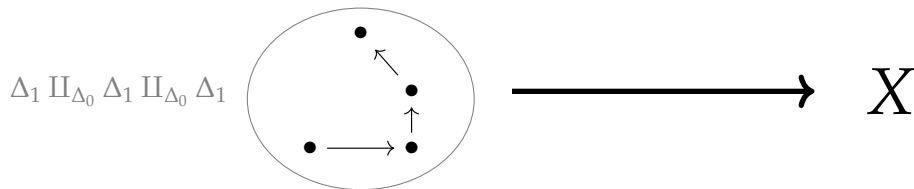
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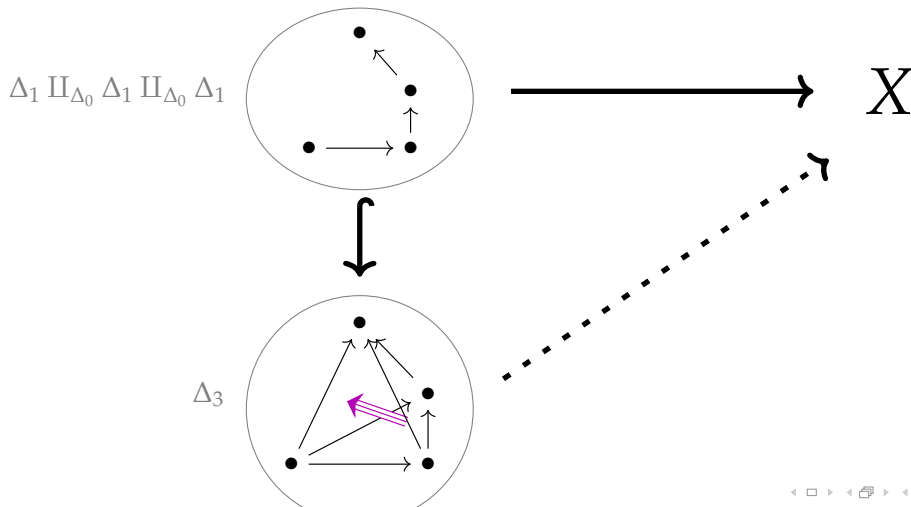
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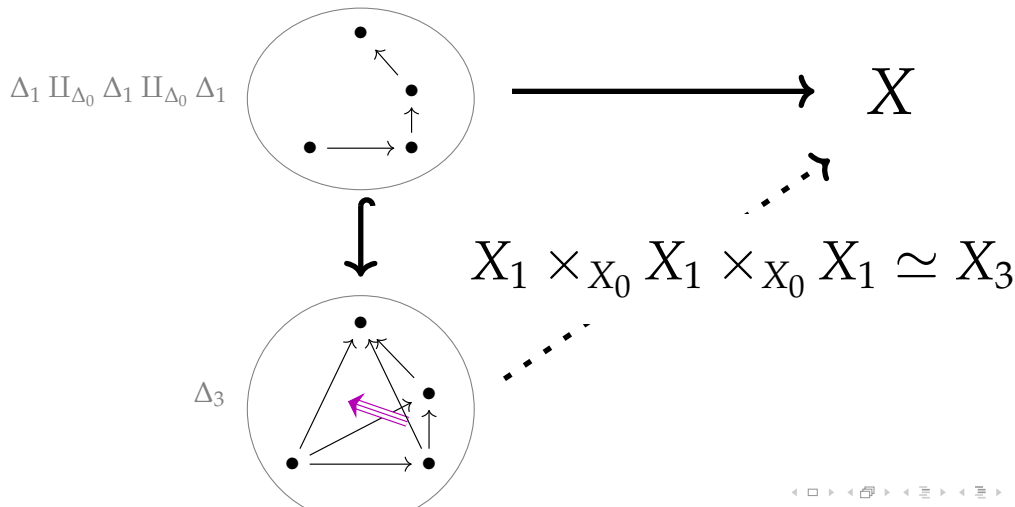
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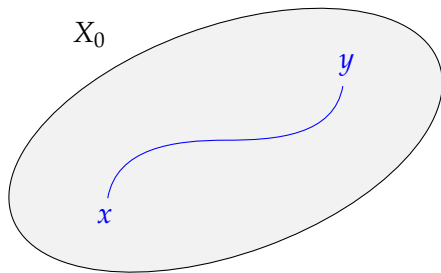
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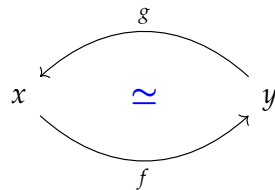
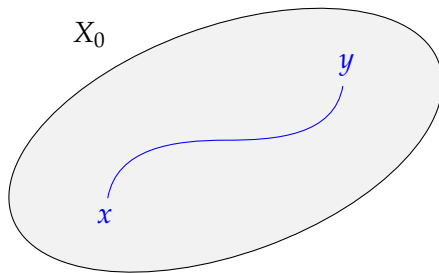
Complete segal spaces

Two coexisting kinds of identifications



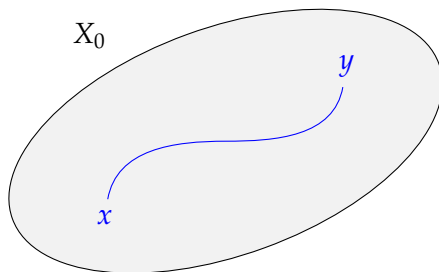
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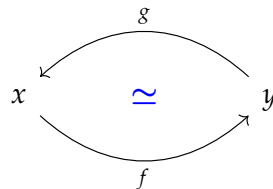


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$\simeq$



Simplicial type theory

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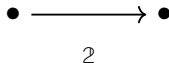
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- Goal: $(\infty, 1)$ -categories = Complete Segal spaces.

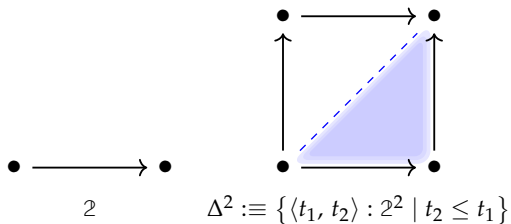
The interval

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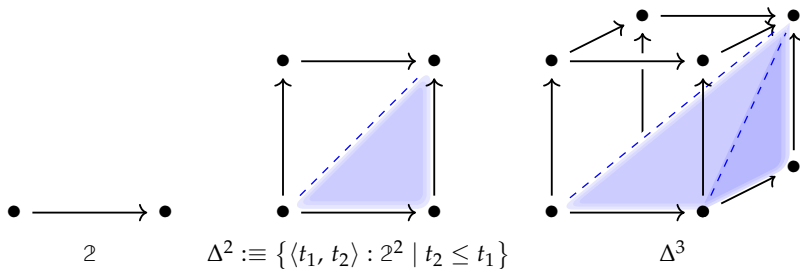
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$\forall n, \quad X_n \equiv \Delta_n \rightarrow X$ is a type

Simplicial types

S shape, and X type:

$S \rightarrow X$ is a type

$$\mathrm{hom}_X(x, y) = \sum_{f:2 \rightarrow X} (f(0) = x) \times (f(1) = y)$$

$\forall n, \quad X_n \equiv \Delta_n \rightarrow X$ is a type

So each type give rise to a *simplicial type* !

Segal types

When does (X_n) satisfies the Segal condition ?

Segal types

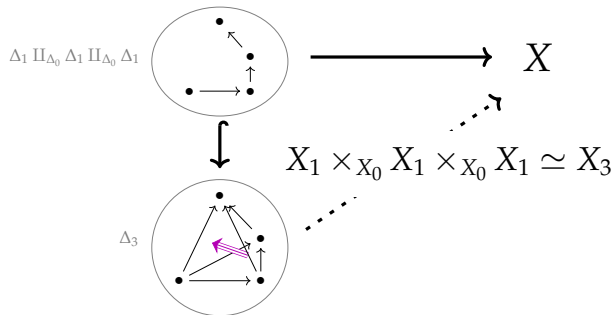
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$$(\Delta^n \rightarrow X) \xrightarrow{\sim} (\Delta_1 \amalg_{\Delta_0} \amalg \cdots \amalg_{\Delta_0} \Delta_1 \rightarrow X)$$

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By horn filling: a Segal type A admits composites $g \circ f$.

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Reminiscent to Rezk-completes/univalent categories in HoTT.

Hom Types in STT

In STT:

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$$x \text{ ————— } y$$

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Then:

$$\mathrm{hom}_{\mathrm{hom}_A(x, y)}(f, g) = \sum_{H: I^2 \rightarrow A} \cdots$$

$$\begin{array}{ccc} x & \text{—} f \text{—} & y \\ \parallel & & \parallel \\ x & \text{—} g \text{—} & y \end{array}$$

The problem of hom-types

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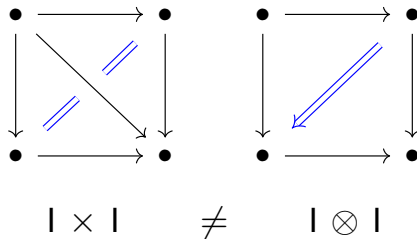
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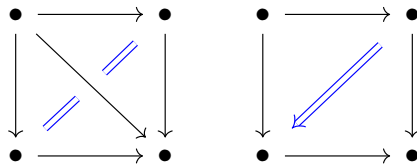


The problem of hom-types

In I^2 : x, y invertible $\Rightarrow (x, y)$ too.

I^2 is 1-categorical: hom of a hom can only be a *groupoid* again.

$$\mathrm{hom}_A(x, y) \neq \sum_{f: I \rightarrow A} (f \cdot 0 = x) \times (f \cdot 1 = y)$$



$$I \times I \neq I \otimes I$$

Replacing $\rightarrow$

- Idea: replacing $(- \times I) \dashv (I \rightarrow -)$ by $(- \otimes I) \dashv (I \multimap -)$

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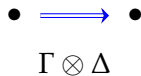
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$$\bullet \xRightarrow{\quad} \bullet$$

$$\Gamma \otimes \Delta$$

$$\bullet \text{ — } \bullet$$

$$\Gamma, \Delta$$

$$\Gamma \otimes \Delta \neq \Delta \otimes \Gamma$$

Interpreted as the Crans-Gray tensor product.

Bunches

$$\Gamma, \Delta ::= \diamond \mid x : A \mid \Gamma \otimes \Delta \mid \Gamma, \Delta$$

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crisp assumptions does not care about functoriality.

Rules for $\rightarrow$ and $\multimap$

$$(\rightarrow\text{-INTRO}) \frac{\Gamma \otimes (x : A) \vdash b : B}{\Gamma \vdash \lambda x. b : A \rightarrow B}$$

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The $\otimes \dashv (\multimap, \rightarrow)$ adjunctions

For crisp types $X, Y, Z :: \mathcal{U}$

$$(X \otimes Y \rightarrow Z) \simeq (X \rightarrow Y \rightarrow Z) \quad (X \otimes Y \multimap Z) \simeq (Y \multimap X \multimap Z)$$

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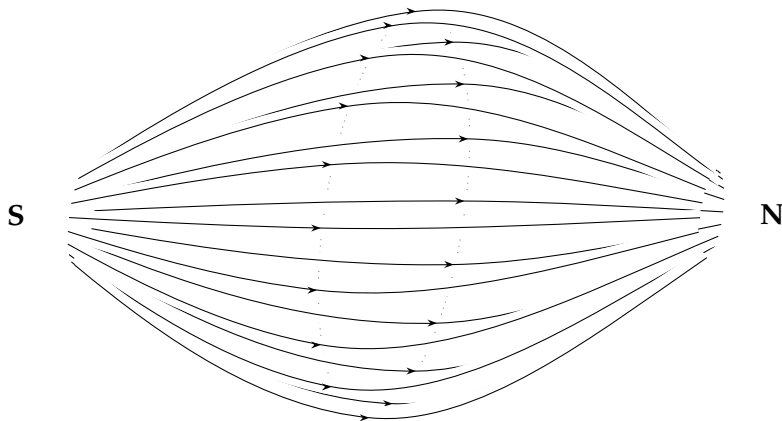
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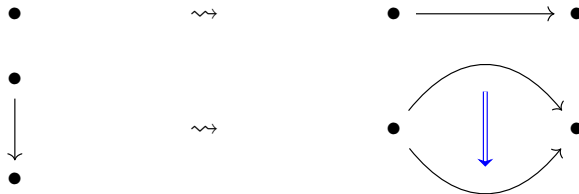
Suspension



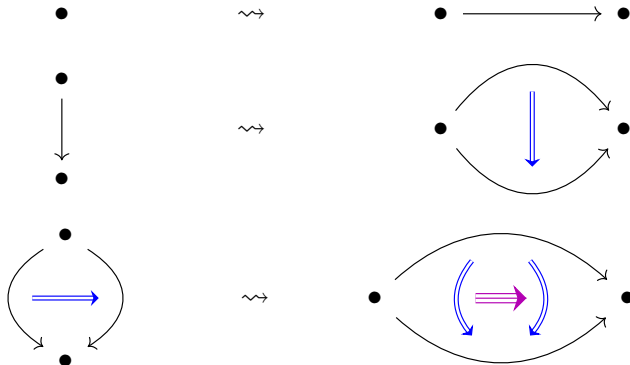
Suspension – Examples



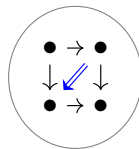
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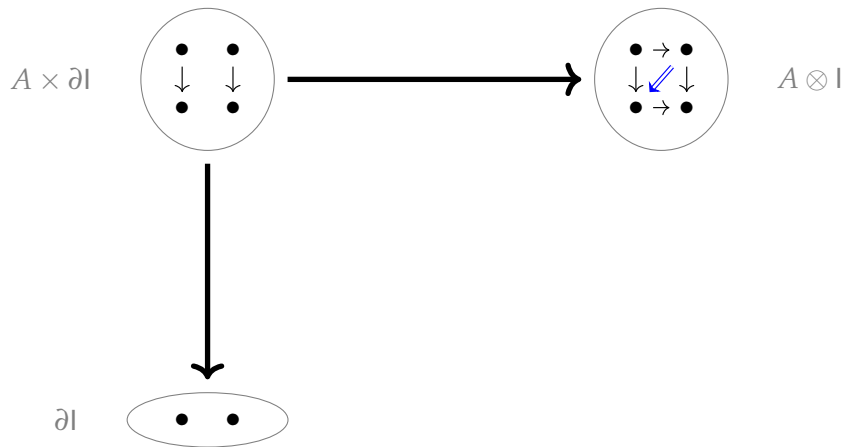


Suspension from $\otimes$

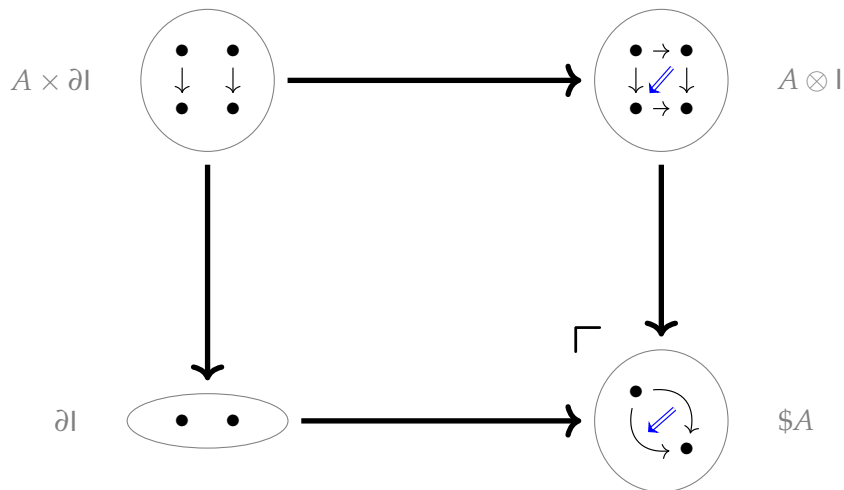


$$A \otimes I$$

Suspension from $\otimes$



Suspension from $\otimes$



Suspension $\dashv$ hom

suspension $\$A$, as a pushout:

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Theorem

$$(\$X \rightarrow Y \text{ sending } 0, 1 \text{ to } y_0, y_1) \simeq (X \rightarrow \mathrm{hom}_Y(y_0, y_1))$$

batt: a type-checker for BaTT

 $/\backslash^{\cdot} _ \cdot^{\cdot} / \backslash$

- Ongoing Ocaml implementation by S. Mimram:
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- Ongoing Ocaml implementation by S. Mimram:
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- Lexer/parser, Type-checker, Metavariables, Unification.
- Small module system, Holes à la Agda, Readable error messages !

Examples

`smimram.github.io/batt/`

The end

Thank you !

Questions ?

